

Kexin Dong

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EDUCATION EXPERIENCE

Massachusetts Institute of Technology (MIT)

Ph.D. in Biology

Cambridge, MA
09/2024–Present

University of Chinese Academy of Sciences (UCAS)

B.S. in Biological Sciences

GPA: 3.97/4.0; Ranking: 1/32 (in major); Honor speaker, Class of 2024

Beijing, China
09/2020-07/2024

Massachusetts Institute of Technology (MIT)

Exchange Undergraduate Student in Biology

GPA: 5.0/5.0

Cambridge, MA
02/2023–05/2023

RESEARCH EXPERIENCE

Francisco J. Sánchez-Rivera Lab & Mike Hemann Lab

Graduate student

Koch Institute at MIT, Cambridge, MA
09/2024–Present

Project 1: Dissecting the molecular mechanisms of clonal expansion and leukemia evolution

- Developing a prime editing-enabled murine HSC platform to model clonal hematopoiesis and quantify CH-AML driver interactions during leukemia evolution.

Project 2: Multiplexed *in vivo* base editing identifies functional gene-variant-context interactions

- Built a multiplexed *in vivo* base-editing screening platform to functionally interrogate endogenous cancer-associated variants in physiological contexts.
- Screened 13,840 gRNAs targeting 7,783 mutations across 489 genes in syngeneic mouse cancer models to map mutation–context–phenotype relationships.
- Revealed context-dependent mutation effects often missed by standard CRISPR-Cas9 nuclease approaches.

Francisco J. Sánchez-Rivera Lab

Undergraduate researcher

Koch Institute at MIT, Cambridge, MA
12/2023-06/2024

Project 1: Computational modeling and evaluation of human variants in mice

- Built a state-of-art Python pipeline, H2M, for precision modeling of human genetic variants in the mouse genome.
- Established a [database](#) containing over 3 million human-mouse variant pairs for the first time via H2M.
- Demonstrated significant relevance of both immunogenicity and pathogenicity between H2M variant pairs by predicting and correlating their NetMHCpan and AlphaMissense scores.

Project 2: Modeling and evaluation of endogenous genetic variants at scale

- Validated remarkable phenotypic diversity of cancer-associated *TP53* variants via a prime editing screen and fluorescence competitive proliferation assays alongside wild-type cells.
- Determined *TP53* variants with high pathogenicity, both well-known and less-studied ones, through comparative analysis of clinical sequencing data and experimental results.

Qianfei Wang Lab

Researcher, Project: Prognostic implications of age in acute myeloid leukemia (AML)

Beijing Institute of Genomics, Beijing, China
08/2022-05/2023

- Performed bioinformatic analysis of age-related mutational patterns in clinical genome data from 2,000 AML patients.
- Uncovered an age specific prognostic implication of *NPM1* mutation, indicating better outcomes in *FLT3-ITD* patients younger than 55.

International Genetically Engineered Machine (iGEM) Project

Group leader, Project: An optimized *in vitro* enzyme system for food decaffeination

UCAS, Beijing, China
04/2021- 11/2021

- Genetically designed, cloned, and purified caffeine dehydrogenase (Cdh), theacrine synthase (CkTcS), and cosubstrates.
- Performed rational redesign via PROSS and directed evolution experiments to improve enzyme thermostability.

TEACHING, LEADERSHIP & SERVICE

Peer Mentor | UCAS Biology Department | Beijing, China

08/2022-07/2023

- Offered weekly academic tutoring and career guidance to 42 first-year Biology undergraduate students.

Teaching Assistant | UCAS Computer Science Department | Beijing, China

02/2022-07/2022

- Held weekly office hours, graded assignments, and exams for over 300 students of *Introduction to Computer Science*.

President | UCAS Student Union, Organization Department | Beijing, China

09/2021-08/2022

- Organized monthly events like movie nights and Science Open Day to enhance campus community engagement.
- Ensured academic and financial support for disadvantaged peers.

Member | UCAS Volunteer Service Team | Beijing, China

09/2020-present

- Biweekly volunteered as COVID Safe Campus Ambassador and Docent of Chinese Paleontological Museum.

PUBLICATION & PRESENTATION

- Gould, S.I., Wuest, A.N., **Dong, K.** et al. High-throughput evaluation of genetic variants with prime editing sensor libraries. Nat Biotechnol (2024). <https://doi.org/10.1038/s41587-024-02172-9>
- **Dong, K.**, Gould, S.I., Li, M. & Sánchez Rivera, F.J. Computational modeling of human genetic variants in mice. bioRxiv (2025). <https://doi.org/10.1101/2025.02.23.639784>
- Acosta, J., Johnson, G.A., Gould, S.I., **Dong, K.** et al. Multiplexed in vivo base editing identifies functional gene-variant-context interactions. bioRxiv (2025). <https://doi.org/10.1101/2025.02.23.639770>

HONORS & AWARDS

- Prize for Open Data, MIT: Selected from 60+ nominations across 25+ departments. 2026
- President Fellowship, MIT: Covered tuition and stipend for the first year of graduate study. 2024
- Excellence Scholarship, UCAS: Top 5% for academic performance. 2023, 2022 & 2021
- Model Student, UCAS: 40 recipients annually for outstanding academic performance and leadership. 2023, 2022 & 2021
- National Scholarship: Top 0.2% of college students in China. 2022
- First Prize, Five-Minute Research Presentation Award: Top 1% of competitors across China. 2022
- Grand Champion, UCAS Undergraduate Research: Funded with ¥20,000 (\$3,000) for research in the Wang Lab. 2022
- Silver Medal, iGEM competition: Led the experimental group. 2021
- Distinguished Leader Award: 8 recipients annually in Beijing, China. 2021

SKILLS

- **Software:** Python, R, Linux Shell, ImageJ, SnapGene, FlowJo, PyMOL, Biorender, Adobe Illustrator
- **Lab Techniques:** Molecular cloning, Plasmid & primer design, Bacteria & mammalian cell culture, genome editing using CRISPR/Cas9, Florescence microscopy, Western blot, Immunostaining, Flow cytometry
- **Language:** English (proficient), Chinese (native)

Updated: October 5, 2026